

MARROWSTONE ADVISORY GROUP | PROGRAM MANAGEMENT OFFICE

Harbor Point Resilience Assessment

Risk Register and Issues Log

Document control	
Document number	HPRA-RISK-001
Version	7
Issued	September 21, 2026
Data as of	September 18, 2026
Next version due	October 5, 2026
Owner	Marisol Vantrease, Program Manager
Contract	CBPA-2026-RA-0417, not-to-exceed \$4,860,000
Distribution	Program leadership, CBPA Client Liaison, CBPA Chief Engineer

Read this first. This register is a snapshot. Every number in it reflects the position on Friday, September 18, 2026. Weekly status notes issued after that date carry more recent figures for inspection progress, cost, and schedule. Where this register and the Project Specification differ on a contractual date, the Specification governs.

1. Summary

This register records the open risks and issues on the Harbor Point Resilience Assessment. It is maintained by the Program Manager, reviewed at each weekly internal meeting, and reissued at least every two weeks as required by Section 10 of the Project Specification.

Program position on September 18, 2026

Measure	Position on September 18, 2026
Assets inspected	10,310 of 12,480 (82.6 percent)
Field inspection forecast finish	October 26, 2026 against a milestone of October 16, 2026 (6 working days late)
Forecast at completion	\$4,954,500 against a not-to-exceed value of \$4,860,000
Forecast overrun	\$94,500 (1.9 percent). Storm costs for the week of September 14 are still being tallied.
Final report (MS-09)	December 18, 2026 (contractual)
Open risks	16 open: 3 Critical, 5 High, 8 Medium, 0 Low. 2 closed since version 6.

What changed since version 6

- Three new risks from the week of Tropical Storm Ferrand: R-014 (final report date), R-015 (dive subcontractor demobilizes), and R-017 (storm damage to the North Breakwater).
- One new Critical risk from a field finding: R-011 (Transformer T-4). It is now the highest scoring risk on the register at 20.
- R-006 (weather closes dive windows) occurred and is logged as issue I-006. Its residual score is 12.
- R-001 and R-004 closed because Pier 4 and the Rail Yard are complete.

Top three open risks by score

Rank	ID	Risk	Score	Owner
1	R-011	Transformer T-4 at Substation Row: critical condition and flood exposure	20	Devin Okonkwo-Hale
2	R-009	Forecast cost at completion exceeds the not-to-exceed value	16	Marisol Vantrease
3	R-014	Final report date is at risk	16	Marisol Vantrease

2. Scoring Method

Each risk is scored as likelihood multiplied by impact, each on a scale of 1 to 5. The bands and handling rules below come from Section 10 of the Project Specification.

Score	Band	Required handling
16 to 25	Critical	Report to the Client Liaison within two business days. Must appear in the weekly status email until the score falls below 16.
12 to 15	High	Mitigation plan with dates. Review weekly.
6 to 11	Medium	Named owner and mitigation. Review every two weeks.

Score	Band	Required handling
1 to 5	Low	Monitor. Review monthly.

Level	Likelihood	Impact
5	Almost certain or already happening	Stops the Program or a terminal, or a contractual failure
4	Likely in the next month	Misses a contractual milestone or exceeds the not-to-exceed value
3	Possible in the next quarter	Delays a milestone by up to 15 working days or costs up to 2 percent
2	Unlikely	Absorbed within float or contingency
1	Rare	Negligible

3. Heat Map

Open risks plotted by likelihood (rows) and impact (columns). Cell color shows the band for that score.

Likelihood	Impact 1	Impact 2	Impact 3	Impact 4	Impact 5
5			R-007		
4			R-008	R-009 R-014	R-011
3			R-010 R-012 R-016	R-006 R-013 R-015	
2			R-002 R-003 R-018	R-017	R-005
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4. Risk Summary

Open risks in descending order of score.

ID	Risk	Category	L	I	Score	Band	Owner	Trend
R-011	Transformer T-4 at Substation Row: critical condition and flood exposure	Asset finding	4	5	20	Critical	Devin Okonkwo-Hale	New
R-009	Forecast cost at completion exceeds the not-to-exceed value	Cost	4	4	16	Critical	Marisol Vantrease	Rising
R-014	Final report date is at risk	Schedule	4	4	16	Critical	Marisol Vantrease	New
R-007	Additional Level II dives exceed the dive budget	Cost	5	3	15	High	Marisol Vantrease	Rising
R-006	Weather closes dive windows at Pier 7 and the Fuel Dock	Schedule	3	4	12	High	Halima Oyelaran-Voss	Occurred
R-008	Energized equipment at Substation Row cannot be inspected without an outage	Schedule	4	3	12	High	Devin Okonkwo-Hale	Rising
R-013	A technical approach does not meet requirement M-07 on data handling	Security and compliance	3	4	12	High	Tomasz Grelling	Steady
R-015	Dive subcontractor demobilizes before the modification is approved	Commercial	3	4	12	High	Marisol Vantrease	New

ID	Risk	Category	L	I	Score	Band	Owner	Trend
R-005	Inspection photographs stored outside the approved repository	Security and compliance	2	5	10	Medium	Tomasz Grelling	Steady
R-010	Confined space entry permits at the Fuel Dock are delayed	Schedule	3	3	9	Medium	Halima Oyelaran-Voss	Steady
R-012	Evaluation panel members are not available for scoring and reconciliation	Resources	3	3	9	Medium	Marisol Vantrease	Steady
R-016	Condition scoring is inconsistent between inspectors	Data quality	3	3	9	Medium	Devin Okonkwo-Hale	Steady
R-017	Storm damage to North Breakwater armor stone after inspection	Data quality	2	4	8	Medium	Devin Okonkwo-Hale	New
R-002	Legacy asset records are incomplete or duplicated	Data quality	2	3	6	Medium	Devin Okonkwo-Hale	Falling
R-003	Inspector certifications lapse during the field phase	Resources	2	3	6	Medium	Halima Oyelaran-Voss	Steady
R-018	Board expectations exceed what early stage cost estimates can support	Stakeholder	2	3	6	Medium	Marisol Vantrease	Steady

5. Risk Detail

One entry for each open risk, in descending order of score.

R-011 Transformer T-4 at Substation Row: critical condition and flood exposure			Score 20 Critical
Category	Asset finding	Owner	Devin Okonkwo-Hale
Opened	September 16, 2026	Likelihood x Impact	4 x 5 = 20
Last reviewed	September 18, 2026	Trend	New
Description	Transformer T-4, installed in 1987, scored condition 1 (Critical) on September 16. Inspectors recorded heavy corrosion at the tank base, an active oil weep at the lower radiator flange, and deteriorated bushing gaskets. The transformer pad sits at 3.5 meters Calder Bay Datum. That is 0.4 meters below the CBPA 2040 design flood elevation of 3.9 meters and 1.1 meters below the 2070 planning scenario of 4.6 meters. In the 2023 flood, water came within 0.2 meters of the pad.		
Consequence	T-4 is the only feed to the reefer racks at Pier 4 and to the transfer pumps at the Fuel Dock. There is no spare transformer on site and the replacement lead time is 38 to 46 weeks. A failure or a flood at T-4 would cut power to refrigerated cargo and stop fuel transfer for months, not days.		
Current controls	Reported to the Chief Engineer on September 17 under the Specification Section 10 rule for a condition 1 asset below the 2040 elevation. CBPA maintenance is checking oil levels weekly.		
Notes	Interim protection is not funded. Design and construction of remediation are outside the Program scope and budget, so any interim barrier needs a CBPA decision and a CBPA funding source. This is the highest scoring risk on the register.		
Mitigation action	Owner	Due	Status
Obtain a quote for an interim flood barrier around T-4	Devin Okonkwo-Hale	September 23	In progress
CBPA to decide funding and owner for interim protection	Dr. Anneliese Thorvaldsen	Not set	Not started
Check whether T-3 can carry part of the T-4 load in an emergency	Devin Okonkwo-Hale	October 2	Planned

R-009 Forecast cost at completion exceeds the not-to-exceed value			Score 16 Critical	
Category	Cost	Owner	Marisol Vantrease	
Opened	September 11, 2026	Likelihood x Impact	4 x 4 = 16	
Last reviewed	September 18, 2026	Trend	Rising	
Description	The forecast at completion is \$4,954,500 against a not-to-exceed value of \$4,860,000. The forecast overrun stands at \$94,500 (1.9 percent), all of it from the additional dive days in R-007. Storm standby and overtime costs for the week of September 14 are still being tallied and are not yet included in this figure.			
Consequence	Any cost above the not-to-exceed value is unrecoverable unless a modification is approved before the cost is incurred. Budget status is Amber under Specification Section 9.3.			
Current controls	Weekly forecast by the Finance Analyst. Modification 03 pending.			
Notes	Expect the forecast overrun to rise in the next version once storm costs are confirmed.			
Mitigation action		Owner	Due	Status
Tally storm standby and overtime costs		Bexley Namgung	September 23	In progress
Identify offsets in Phase 4 travel and workshops		Marisol Vantrease	September 30	Planned

R-014 Final report date is at risk			Score 16 Critical	
Category	Schedule	Owner	Marisol Vantrease	
Opened	September 18, 2026	Likelihood x Impact	4 x 4 = 16	
Last reviewed	September 18, 2026	Trend	New	
Description	The contractual date for the final resilience report and remediation roadmap is December 18, 2026 (MS-09). Field inspection is now forecast to finish on October 26 instead of October 16. That compresses the time to produce the draft report (MS-08, November 20) from 25 working days to 19.			
Consequence	A late or thin final report weakens the Board briefing and is a contractual miss.			
Current controls	Recovery options are being prepared for the October 8 program review.			
Notes	A move to January 15, 2027 was proposed internally on September 18. It has not been discussed with CBPA and no modification has been drafted. December 18, 2026 remains the contractual date and the date used for reporting.			
Mitigation action		Owner	Due	Status
Prepare recovery options, including early start on modeling with partial data		Marisol Vantrease	October 6	Planned
Raise the report date with the Client Liaison		Marisol Vantrease	September 24	Planned

R-007 Additional Level II dives exceed the dive budget			Score 15 High	
Category	Cost	Owner	Marisol Vantrease	
Opened	August 21, 2026	Likelihood x Impact	5 x 3 = 15	
Last reviewed	September 18, 2026	Trend	Rising	
Description	Level I inspection found significant section loss on 212 steel H-piles in bents 14 to 22 at Pier 7. Requirement M-09 requires the Level II sample to increase where Level I finds significant damage. Greywater Subsea Services estimates 14 additional dive days, taking the total from 46 to 60.			
Consequence	At \$6,750 per dive day the additional scope costs \$94,500, which is not in the budget.			
Current controls	Contract Modification 03 was submitted to CBPA on September 10. Work is proceeding at Marrowstone's risk for up to five dive days.			
Notes	Likelihood is 5 because the additional dives are required by the Specification. The open question is who pays.			
Mitigation action		Owner	Due	Status
Track Modification 03 with the Contracting Officer		Tomasz Grelling	Weekly	In progress
Plain language note on pile findings for the Client Liaison		Devin Okonkwo-Hale	September 15	Done

R-006 Weather closes dive windows at Pier 7 and the Fuel Dock			Score 12 High	
Category	Schedule	Owner	Halima Oyelaran-Voss	
Opened	June 19, 2026	Likelihood x Impact	3 x 4 = 12	
Last reviewed	September 18, 2026	Trend	Occurred	
Description	Underwater work stops when a tropical storm watch is issued for Calder Bay or when sea state or visibility exceed limits. Storm season runs through November 30.			
Consequence	Each lost dive day pushes out completion of Pier 7 and adds standby cost. The Specification assumes no more than 10 working days lost to weather in Phase 2.			
Current controls	Daily marine forecast review. Land-based work is resequenced when dives stop.			
Notes	This risk occurred during the week of September 14 (issue I-006). Five working days were lost. The score shown is the residual risk of a second event.			
Mitigation action		Owner	Due	Status
Recovery plan for five lost dive days		Halima Oyelaran-Voss	September 23	In progress
Agree a weekend dive option with Harbor Operations		Rourke Castellanos	September 25	Planned

R-008 Energized equipment at Substation Row cannot be inspected without an outage			Score 12 High	
Category	Schedule	Owner	Devin Okonkwo-Hale	
Opened	June 19, 2026	Likelihood x Impact	4 x 3 = 12	
Last reviewed	September 18, 2026	Trend	Rising	
Description	External inspection at Substation Row is 70 percent complete. Of the 468 assets remaining, 234 are internal components of energized switchgear and transformers that can only be inspected during a planned outage. Outages are allowed only on Saturdays from 6:00 AM to 12:00 PM and need Chief Engineer approval five business days ahead.			
Consequence	Without an approved outage, requirement M-01 (100 percent of assets) cannot be met and field inspection cannot close.			
Current controls	Outage request in preparation with two candidate dates.			
Notes	Candidate dates under discussion are Saturday, October 10 and Saturday, October 17.			
Mitigation action		Owner	Due	Status
Ask the Chief Engineer for candidate outage dates		Marisol Vantrease	September 16	Done
Submit formal outage request		Devin Okonkwo-Hale	September 28	Planned

R-013 A technical approach does not meet requirement M-07 on data handling			Score 12 High	
Category	Security and compliance	Owner	Tomasz Grelling	
Opened	September 11, 2026	Likelihood x Impact	3 x 4 = 12	
Last reviewed	September 18, 2026	Trend	Steady	
Description	Requirement M-07 is pass or fail. One submission proposes to host its model in a way that may not meet the requirement for United States based environments authorized for Controlled Unclassified Information. Clarification request CR-02 was issued on September 11 and answered on September 17.			
Consequence	A non-compliant approach cannot be recommended whatever its score. If the highest scoring approach fails M-07, the recommendation changes.			
Current controls	Compliance and security lead reviews each submission against M-06, M-07, and M-08 before scores are reconciled.			
Mitigation action		Owner	Due	Status
Assess the CR-02 response against M-07		Tomasz Grelling	September 23	In progress

R-015 Dive subcontractor demobilizes before the modification is approved			Score 12 High	
Category	Commercial	Owner	Marisol Vantrease	
Opened	September 18, 2026	Likelihood x Impact	3 x 4 = 12	
Last reviewed	September 18, 2026	Trend	New	
Description	Greywater Subsea Services has another client asking for the same dive crew from early October. Marrowstone has authorized work at risk for five dive days beyond the original 46.			
Consequence	If the crew leaves, the remaining underwater inspection at Pier 7 stops and requirement M-01 cannot be met on any schedule currently under discussion.			
Current controls	Weekly call with Greywater. Modification 03 is being tracked with the Contracting Officer.			
Mitigation action		Owner	Due	Status
Confirm crew availability through October		Halima Oyelaran-Voss	September 22	In progress
Set a decision deadline for Modification 03 with CBPA		Marisol Vantrease	September 24	Planned

R-005 Inspection photographs stored outside the approved repository			Score 10 Medium	
Category	Security and compliance	Owner	Tomasz Grelling	
Opened	June 8, 2026	Likelihood x Impact	2 x 5 = 10	
Last reviewed	September 18, 2026	Trend	Steady	
Description	Field tablets capture several hundred photographs of port infrastructure each day. Requirement M-07 and Specification Section 11 require that all Program data, including photographs, stay inside the approved Program repository.			
Consequence	Sensitive images of critical infrastructure held outside an authorized environment would be a reportable incident and could suspend field work.			
Current controls	Managed tablets with personal cloud sync blocked. Daily upload check. One near miss on August 19 (issue I-003) was contained within the hour and the device policy was tightened.			
Notes	Likelihood held at 2 after the August near miss because the policy change closed the specific gap.			
Mitigation action		Owner	Due	Status
Monthly audit of tablet configuration		Tomasz Grelling	October 1	Planned
Refresher briefing for all inspectors		Halima Oyelaran-Voss	September 28	Planned

R-010 Confined space entry permits at the Fuel Dock are delayed			Score 9 Medium	
Category	Schedule	Owner	Halima Oyelaran-Voss	
Opened	July 31, 2026	Likelihood x Impact	3 x 3 = 9	
Last reviewed	September 18, 2026	Trend	Steady	
Description	Each tank entry at the Fuel Dock and Tank Farm needs its own CBPA permit. Permits for Tanks 1 and 2 took nine working days to issue. Tanks 3 to 6 are still to be entered.			
Consequence	A slow permit holds up roughly 90 assets per tank.			
Current controls	Permit requests are now submitted three weeks ahead.			
Mitigation action		Owner	Due	Status
Permits for Tanks 3 and 4		Halima Oyelaran-Voss	September 25	In progress
Permits for Tanks 5 and 6		Halima Oyelaran-Voss	October 2	Planned

R-012 Evaluation panel members are not available for scoring and reconciliation			Score 9 Medium	
Category	Resources	Owner	Marisol Vantrease	
Opened	August 21, 2026	Likelihood x Impact	3 x 3 = 9	
Last reviewed	September 18, 2026	Trend	Steady	
Description	The five voting panel members all have operational roles. Three of five had submitted Technical Merit scores by September 18.			
Consequence	Late scores delay the reconciliation meeting and put the October 23 recommendation memo (MS-06) at risk.			
Current controls	Scoring deadlines are in calendars. The Program Manager follows up individually.			
Mitigation action		Owner	Due	Status
Collect remaining individual scores		Marisol Vantrease	October 2	In progress
Hold reconciliation meeting		Marisol Vantrease	October 6	Planned

R-016 Condition scoring is inconsistent between inspectors			Score 9 Medium	
Category	Data quality	Owner	Devin Okonkwo-Hale	
Opened	July 10, 2026	Likelihood x Impact	3 x 3 = 9	
Last reviewed	September 18, 2026	Trend	Steady	
Description	The third quality audit found that 4 of 50 sampled records differed from the auditor's score by more than one point. All four were culverts or drainage assets.			
Consequence	Inconsistent scores undermine the risk tiers and the remediation roadmap.			
Current controls	Audit of 50 records every three weeks. Scoring guidance for drainage assets was tightened in July.			
Mitigation action		Owner	Due	Status
Recalibration session for all inspectors		Devin Okonkwo-Hale	September 29	Planned
Re-score all Rail Yard culverts inspected before July 31		Halima Oyelaran-Voss	October 9	Planned

R-017 Storm damage to North Breakwater armor stone after inspection			Score 8 Medium	
Category	Data quality	Owner	Devin Okonkwo-Hale	
Opened	September 18, 2026	Likelihood x Impact	2 x 4 = 8	
Last reviewed	September 18, 2026	Trend	New	
Description	The North Breakwater inspection was completed on September 12, two days before Tropical Storm Ferrand. Wave action during the storm may have displaced armor stone in 38 sections on the exposed face.			
Consequence	Condition scores recorded before the storm may no longer be accurate for those sections.			
Current controls	Post-storm recheck of the 38 sections is scheduled. The sections remain counted as inspected.			
Mitigation action		Owner	Due	Status
Recheck 38 armor stone sections		Halima Oyelaran-Voss	September 30	Planned

R-002 Legacy asset records are incomplete or duplicated			Score 6 Medium	
Category	Data quality	Owner	Devin Okonkwo-Hale	
Opened	June 12, 2026	Likelihood x Impact	2 x 3 = 6	
Last reviewed	September 18, 2026	Trend	Falling	
Description	The CBPA legacy asset system held 13,104 records against a reconciled baseline of 12,480. Duplicate and retired records continue to surface during field work, most recently 61 duplicate asset IDs in the Rail Yard records.			
Consequence	Inspectors waste time looking for assets that do not exist, and the inspection dataset may not reconcile cleanly to the baseline at acceptance (requirement M-01).			
Current controls	Baseline fixed at 12,480 on June 26. Every duplicate found in the field is logged and reconciled weekly.			
Notes	No further duplicates found since September 11.			
Mitigation action		Owner	Due	Status
Reconcile 61 Rail Yard duplicates		Devin Okonkwo-Hale	September 11	Done
Final reconciliation of dataset to baseline before DL-04		Devin Okonkwo-Hale	October 20	Planned

R-003 Inspector certifications lapse during the field phase			Score 6 Medium	
Category	Resources	Owner	Halima Oyelaran-Voss	
Opened	June 12, 2026	Likelihood x Impact	2 x 3 = 6	
Last reviewed	September 18, 2026	Trend	Steady	
Description	Two of the twelve inspectors hold certifications that expire on October 9 and October 14, inside the field inspection period.			
Consequence	An inspector with a lapsed certification cannot sign inspection records. Losing two inspectors for a week would cost roughly 400 inspections.			
Current controls	Renewal sessions are booked for both inspectors on October 6.			
Notes	Becomes more important now that the field phase is forecast to run past October 16.			
Mitigation action		Owner	Due	Status
Confirm renewal bookings		Halima Oyelaran-Voss	September 25	In progress
Identify backup inspector from the Marrowstone bench		Halima Oyelaran-Voss	September 30	Planned

R-018 Board expectations exceed what early stage cost estimates can support			Score 6 Medium	
Category	Stakeholder	Owner	Marisol Vantrease	
Opened	August 14, 2026	Likelihood x Impact	2 x 3 = 6	
Last reviewed	September 18, 2026	Trend	Steady	
Description	The roadmap will carry Class 4 cost estimates with an accuracy range of minus 30 percent to plus 50 percent. Some Commissioners have said they expect numbers they can budget against.			
Consequence	The Board may see the roadmap as not actionable.			
Current controls	The Client Liaison is briefing Commissioners on what a Class 4 estimate means.			
Mitigation action		Owner	Due	Status
One page explainer on estimate classes for the Board package		Marisol Vantrease	November 13	Planned

6. Closed Risks

ID	Risk	Owner	Closed	Reason
R-001	Berth access windows at Pier 4 restrict inspection	Halima Oyelaran-Voss	September 14, 2026	Pier 4 inspection complete on September 9.
R-004	Rail Yard track possessions conflict with rail operations	Halima Oyelaran-Voss	September 18, 2026	Rail Yard inspection complete on September 17.

7. Issues Log

An issue is a risk that has happened. Open issues are reviewed at every weekly meeting.

ID	Issue	Raised	Owner	Status	Resolution or next step
I-001	Vessel overstayed at Pier 4 berth 2. Two mornings of inspection lost.	July 15, 2026	Halima Oyelaran-Voss	Closed	Standby task list agreed July 21. Time recovered by July 31.
I-002	Three inspection records missing photographs in the first quality audit.	July 30, 2026	Devin Okonkwo-Hale	Closed	Records corrected July 31.
I-003	A field tablet synced 14 photographs to a personal cloud account.	August 19, 2026	Tomasz Grelling	Closed	Contained and deleted within one hour. Reported to CBPA. Device policy tightened August 20.
I-004	61 duplicate asset IDs found in Rail Yard legacy records.	September 2, 2026	Devin Okonkwo-Hale	Closed	Reconciled September 11.
I-005	Level I dives found significant section loss on 212 piles in bents 14 to 22 at Pier 7.	September 9, 2026	Devin Okonkwo-Hale	Open	Level II sample increased by 14 dive days. Modification 03 submitted September 10.
I-006	Tropical Storm Ferrand stopped dive operations from September 14 to September 18. Five working days lost.	September 14, 2026	Halima Oyelaran-Voss	Open	Recovery plan due September 23. Standby and overtime costs being tallied.
I-007	The Client Liaison learned of the dive stoppage from the Harbor Operations Center before the Program told her.	September 17, 2026	Marisol Vantrease	Closed	Same-day notice rule adopted September 17 for any work stoppage.

8. Version History

Version	Issued	Summary of change
1	June 12, 2026	Initial register: R-001 to R-005.
2	July 3, 2026	Added R-006 and R-008. Updated after the inventory baseline.
3	July 31, 2026	Added R-010 and R-016. First field phase update.
4	August 14, 2026	Added R-018 after the first monthly program review with inspection data.

Version	Issued	Summary of change
5	August 28, 2026	Added R-007 and R-012. No risk above 12.
6	September 11, 2026	Added R-009 and R-013. R-007 raised to 15.
7	September 21, 2026	Added R-011, R-014, R-015, R-017. R-006 occurred. Closed R-001 and R-004.

The next version is due October 5, 2026 and will include confirmed storm costs, the outcome of Modification 03, and the result of the Substation Row outage request.